# 符號定義

- $k$= 第幾個灰階 (which gray level)
- $r_k$：原始灰階值 (original gray level)
- $n_k$：該灰階出現的次數 (number of pixels with gray level $r_k$)
- $N$：總像素數 (total number of pixels)
- $PMF(r_k)$：機率質量函數 $=\frac{n_k}{N}$ (probability of gray level $r_k$)
- $CDF(r_k)$：累積分布函數 $=\sum_{i=1}^{k} PMF(r_i)$ (cumulative probability)
- $r_{\max}, r_{\min}$：灰階最大值與最小值 (maximum and minimum gray level)
- $s_k$：等化後的新灰階值 (new gray level after equalization)
- $s_k = round(CDF(r_k) \times (r_{\max} - r_{\min}) + r_{\min})$

# EX1:直方圖等化(Histogram Equalization)

- 一張4階的3×3=9個像素的灰階影像
- 運用直方圖等化(HE)，找出轉換後的像素值s=T(r)。
- A 4-level grayscale image with $3 \times 3 = 9$ pixels is given.
- Find the equalized values $s = T(r)$.

- $r_1 = 0$ ,$r_2 = 1$ ,$r_3 = 2$ ,$r_4 = 3$
- $n_1 = 4$ ,$n_2 = 2$ ,$n_3 = 2$ ,$n_4 = 1$

$r_1$：總共有4個
(number of pixels = 4)

| | | |
|---|---|---|
| $r_4$ | $r_1$ | $r_3$ |
| $r_1$ | $r_2$ | $r_3$ |
| $r_1$ | $r_2$ | $r_1$ |

| | | |
|---|---|---|
| 3 | 0 | 2 |
| 0 | 1 | 2 |
| 0 | 1 | 0 |

# EX1:直方圖等化(HE) (1/3) 算PMF

- 一張4階的3×3=9個像素的灰階影像
- 運用直方圖等化(HE)，找出轉換後的像素值s=T(r)。
- A 4-level grayscale image with 3 × 3 = 9 pixels is given.
- Find the equalized values $s = T(r)$.

$$PMF(r_1) = \frac{n_1}{\mathrm{N}} = \frac{4}{9}$$

$$PMF(r_2) = \frac{n_2}{\mathrm{N}} = \frac{2}{9}$$

$$PMF(r_3) = \frac{n_3}{\mathrm{N}} = \frac{2}{9}$$

$$PMF(r_4) = \frac{n_4}{\mathrm{N}} = \frac{1}{9}$$

- $n_1 = 4$ ,$n_2 = 2$ ,$n_3 = 2$ ,$n_4 = 1$

會使用到的公式：

$$PMF(r_k) = \frac{n_k}{\mathrm{N}}$$

# EX1:直方圖等化(HE) (2/3) 算CDF

$$CDF(r_1) = \frac{n_1}{N} = \frac{4}{9}$$

$$CDF(r_2) = \frac{n_1}{N} + \frac{n_2}{N} = \frac{6}{9}$$

$$CDF(r_3) = \frac{n_1}{N} + \frac{n_2}{N} + \frac{n_3}{N} = \frac{8}{9}$$

$$CDF(r_4) = \frac{n_1}{N} + \frac{n_2}{N} + \frac{n_3}{N} + \frac{n_4}{N} = \frac{9}{9}$$

$$PMF(r_1) = \frac{n_1}{N} = \frac{4}{9}$$

$$PMF(r_2) = \frac{n_2}{N} = \frac{2}{9}$$

$$PMF(r_3) = \frac{n_3}{N} = \frac{2}{9}$$

$$PMF(r_4) = \frac{n_4}{N} = \frac{1}{9}$$

會使用到的公式:

$$CDF(r_k) = \sum_{i=1}^{k} PMF(r_i)$$

# EX1:直方圖等化(HE) (3/3) 算s=T(r)

$$s_k = round(CDF(r_k) \times (r_{max} - r_{min}) + s_{min})$$

$$s_1 = round(CDF(r_1) \times (3-0)) = round\left(\frac{4}{9} \times 3\right) = round(1.33) \approx 1$$

$$s_2 = round(CDF(r_2) \times (3-0)) = round\left(\frac{6}{9} \times 3\right) = round(2) \approx 2$$

$$s_3 = round(CDF(r_3) \times (3-0)) = round\left(\frac{8}{9} \times 3\right) = round(2.66) \approx 3$$

$$s_4 = round(CDF(r_4) \times (3-0)) = round\left(\frac{9}{9} \times 3\right) = round(3) \approx 3$$

$$CDF(r_1) = \frac{n_1}{N} = \frac{4}{9}$$

$$CDF(r_2) = \frac{n_1}{N} + \frac{n_2}{N} = \frac{6}{9}$$

$$CDF(r_3) = \frac{n_1}{N} + \frac{n_2}{N} + \frac{n_3}{N} = \frac{8}{9}$$

$$CDF(r_4) = \frac{n_1}{N} + \frac{n_2}{N} + \frac{n_3}{N} + \frac{n_4}{N} = \frac{9}{9}$$

| | $k$=1 | $k$=2 | $k$=3 | $k$=4 |
|---|---|---|---|---|
| 原始強度 $r_k$ | 0 | 1 | 2 | 3 |
| 轉換後強度 $s_k$ | 1 | 2 | 3 | 3 |

#

會使用到的公式：

$$s_k = round(CDF(r_k) \times (r_{max} - r_{min}) + s_{min})$$

# Extra Question

- 把算好的等化後灰階值 $s_k$ 填回原始影像。
- Fill the equalized gray values $s_k$ back into the original image.

| | $k$=1 | $k$=2 | $k$=3 | $k$=4 |
|---|---|---|---|---|
| 原始強度 $r_k$ | 0 | 1 | 2 | 3 |
| 轉換後強度 $s_k$ | 1 | 2 | 3 | 3 |

Before HE

| | | |
|---|---|---|
| 3 | 0 | 2 |
| 0 | 1 | 2 |
| 0 | 1 | 0 |

After HE

| | | |
|---|---|---|
| 3 | 1 | 3 |
| 1 | 2 | 3 |
| 1 | 2 | 1 |

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